Exp no :14 Experiment to understand Feature Scaling

Program:

```
import numpy as np
import pandas as pd

df = pd.read_csv('pre-
process_datasample.csv')
print("Original Dataset:") print(df) print()

df['Country'] = df.Country.fillna(df.Country.mode()[0])

features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

from sklearn.impute import SimpleImputer age_imputer =
SimpleImputer(strategy="mean", missing_values=np.nan)
salary_imputer = SimpleImputer(strategy="mean",
missing_values=np.nan) age_imputer.fit(features[:, [1]])
salary_imputer.fit(features[:, [2]]) features[:, [1]] =
age_imputer.transform(features[:, [1]]) features[:, [2]] =
salary_imputer.transform(features[:, [2]]) print("Features after
handling missing values:") print(features) print()

from sklearn.preprocessing import
OneHotEncoder oh =
OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:,
[0]]) print("One-Hot Encoded Country:")
print(Country) print()

final_set = np.concatenate((Country, features[:, [1, 2]]),
axis=1) print("Final feature set (before scaling):")
print(final_set) print()

from sklearn.preprocessing import
StandardScaler sc = StandardScaler()
sc.fit(final_set) feat_standard_scaler =
sc.transform(final_set) print("Feature set
after Standard Scaling:")
print(feat_standard_scaler) print()

from sklearn.preprocessing import
MinMaxScaler mms =
MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set) feat_minmax_scaler =
mms.transform(final_set) print("Feature
set after Min-Max Scaling:")
print(feat_minmax_scaler)
```

Output:

Original Dataset:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Features after handling missing values:

```
[['France' 44.0 72000.0]
 ['Spain' 27.0 48000.0]
 ['Germany' 30.0 54000.0]
 ['Spain' 38.0 61000.0]
 ['Germany' 40.0 63777.77777777778]
 ['France' 35.0 58000.0]
 ['Spain' 38.77777777777778 52000.0]
 ['France' 48.0 79000.0]
 ['Germany' 50.0 83000.0]
 ['France' 37.0 67000.0]]
```

One-Hot Encoded Country:

```
[[1. 0. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]
[0.  1. 0.]
[1.  0. 0.]
[0.  0. 1.]
[1.  0. 0.]
[0.  1. 0.]
[1.  0. 0.]]
```

Final feature set (before scaling):

```
[[1.0 0.0 0.0 44.0 72000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [0.0 1.0 0.0 30.0 54000.0]
 [0.0 0.0 1.0 38.0 61000.0]
 [0.0 1.0 0.0 40.0 63777.77777777778]
 [1.0 0.0 0.0 35.0 58000.0]
 [0.0 0.0 1.0 38.77777777777778 52000.0]
 [1.0 0.0 0.0 48.0 79000.0]
```

```
 [0.0 1.0 0.0 50.0 83000.0]
 [1.0 0.0 0.0 37.0 67000.0]]

Feature set after Standard Scaling:
[[ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01
7.58874362e-01
   7.49473254e-01]
 [-8.16496581e-01 -6.54653671e-01    1.52752523e+00 -
1.71150388e+00
  -1.43817841e+00]
 [-8.16496581e-0   1.52752523e+00 -6.54653671e-01 -
1.27555478e+00
  -8.91265492e-01]
 [-8.16496581e-01 -6.54653671e-01   1.52752523e+00 -
1.13023841e-01
  -2.53200424e-01]
 [-8.16496581e-01   1.52752523e+00 -6.54653671e-01
1.77608893e-01
   6.63219199e-16]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -
5.48972942e-01
  -5.26656882e-01]
 [-8.16496581e-01 -6.54653671e-01   1.52752523e+00
0.00000000e+00
  -1.07356980e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01
1.34013983e+00
   1.38753832e+00]
 [-8.16496581e-01   1.52752523e+00 -6.54653671e-01
1.63077256e+00
   1.75214693e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -
2.58340208e-01
   2.93712492e-01]]

Feature set after Min-Max Scaling:
[[1.        0.        0.        0.73913043 0.68571429]
 [0.        0.        1.        0.        0.        1
 [0.        1.        0.        0.13043478 0.17142857]
 [0.        0.        1.        0.47826087 0.37142857]
                1.        0.        0.56521739 0.45079365]
[0.            0.        0.        0.34782609 0.28571429]
[0.            0.        1.        0.51207729 0.11428571]
[1.            0.        0.        0.91304348 0.88571429]
[0.            1.        0.        1.        1.        1
[1.            0.        0.        0.43478261 0.54285714]]
```